

KANNAN APPIYA SANTHARAM

Senior Lead Software Engineer (Frontend/Full Stack) | React | TypeScript | Node.js | AI-Native

Chennai, India

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Nationality: Indian | **Notice Period:** 60 Days

PROFESSIONAL SUMMARY

10.5+ years in enterprise SaaS, building across React, TypeScript and Node.js. Currently leading an engineering team on a platform managing millions of customer endpoints, owning frontend architecture across a 12-package monorepo plus design systems, build and runtime performance, alongside AI-native engineering with LLM orchestration and MCP servers. Seeking a Senior Lead / Staff Engineer or Engineering Manager role.

KEY ACHIEVEMENTS

- **Lead a frontend team of 6** at SuperOps on a platform managing millions of endpoints, shipping weekly and owning hiring, performance reviews, architecture and roadmap prioritisation with product.
- **Enabled 50+ enterprise accounts** at Freshworks, \$100K to \$1M ARR each, with custom Node.js middleware and microservices that met requirements the core product could not.
- **Delivered custom features and public apps** across 4 Freshworks products to 10,000+ customers, built in React with Node.js services on AWS, secured with JWT and, for public apps, OAuth 2.0 where the third-party API supported it.
- **Architected an AI-driven end-to-end testing platform** that cut full product regression from 6 hours of manual testing to a 40-minute automated run.
- **Built a Python MCP server** exposing the codebase as a queryable graph, shipped as a Docker image on AWS EC2 behind token auth, now used by 3 engineering squads for their AI tooling.
- **Cut product initial load time by 70%**, from 5 seconds to 1.5 seconds.
- **Hired 9 engineers** and mentored 10+ across SuperOps and Freshworks, at 90% retention.

SKILLS

Frontend & Full Stack: React, Next.js, TypeScript, Node.js, Python, Microservices, GraphQL, REST API Design, JWT & OAuth 2.0, MySQL, DynamoDB, Tailwind CSS, Playwright, Jest & Vitest

Platform & Performance: Monorepos & Module Federation, Rspack & Webpack 5, Micro-Frontends, Design Systems, Core Web Vitals, Jenkins CI/CD, Docker, AWS

AI-Native Engineering: LLM Orchestration, Model Context Protocol (MCP) Servers, Agent Skills & Guardrails, RAG & Embeddings, Amazon Bedrock AgentCore, Claude Code

Leadership: Team Leadership (6 engineers), Hiring & Mentorship (10+ engineers), Solution Architecture & System Design, Stakeholder & Customer Management

WORK EXPERIENCE

SUPEROPS

Senior Lead Software Engineer

Chennai, India

Jul 2022 – Present

Progression: Senior Software Engineer to Lead Software Engineer to Senior Lead Software Engineer

- Hands-on delivery of product features in React and TypeScript, including role-based access control (RBAC) and near-real-time operational dashboards tuned with custom cached Apollo GraphQL hooks to balance data freshness against network load, in partnership with product and UI/UX teams.
- Manage the team's sprint planning, architecture reviews, interviews, mentoring and performance feedback, and roadmap prioritisation with product managers.
- Reduced dev build time by 96%, from 2 minutes to 5 seconds, through a solo Webpack 5 to Rspack migration delivered in 3 weeks across 12 monorepo packages, reclaiming roughly 50 engineer-hours a month across a 15-engineer frontend org and accelerating time-to-market downstream.

- Architected the AI-driven end-to-end testing platform in React and Node.js, where LLM agents generate, run and self-heal a 300+ test-case Playwright suite that now gates every release in Jenkins CI/CD.
- Improved developer productivity by integrating custom Claude Code skills and MCP servers into the production development workflow, with multi-tier write-scope guardrails that let engineers run autonomous agents without exposing core platform interfaces to risk.
- Built a Python MCP server exposing the frontend codebase as a queryable graph, packaged as a Docker image and hosted on AWS EC2 behind token-based authentication, now the shared context source for 3 engineering squads' AI tooling.
- Built a headless Base UI design system of 30+ reusable components now used across 40+ pages and sections, with customised design tokens and a Claude Code skill that generates production components from a Figma node link via MCP, cutting page development time from 2 days, even with AI assistance, down to a few hours.
- Cut the product's initial page load from 5 seconds to 1.5 seconds through route-level code splitting, React lazy loading and per-package bundle chunking, ending the slowness-related support tickets, and removed 30,000+ lines of dead code and unused dependencies.

FRESHWORKS

Senior Software Engineer (Customer-Facing Engineering)

Chennai, India

Jun 2018 – Jul 2022

Progression: Onboarding Engineer to Senior Software Engineer

- Technical point of contact for 100+ mid-market and enterprise accounts over four years, partnering with Support, Customer Success and Sales to turn account requirements into shipped solutions, recognised with the Customer Champion Award.
- Built tier-1 SaaS integrations over REST API services, including the Jira integration adopted by 4,000+ Freshworks customers, alongside Salesforce, Zendesk and ServiceNow integrations for those accounts, expanding platform fit in enterprise deals.
- Shipped custom features and public apps in React and Node.js across 4 Freshworks products to 10,000+ customers, with JWT for app authentication and OAuth 2.0 where the third-party API supported it.
- Built the Node.js middleware and microservices behind 50+ enterprise accounts worth \$100K to \$1M ARR each, deployed on AWS EC2 with S3, DynamoDB and MySQL, delivering requirements the core product could not and keeping onboarding and renewals on track.
- Mentored 7 junior and onboarding engineers on React, REST API development and UI architecture through structured code and design reviews.

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Software Developer (acquired by Freshworks)

Chennai & Bengaluru, India

Mar 2016 – Jun 2018

Progression: Web Application Developer & Designer to Software Developer

- Built an admin dashboard that listed the features, bandwidth and storage limits behind each Stripe plan and let admins create custom plans on demand, increasing self-serve revenue.
- Built video player customisation from scratch with configurable controls, annotations and call-to-action overlays, increasing viewer conversion.
- Built responsive, high-converting company and product landing pages.
- Integrated Freshservice with DocuSign over REST API services, unlocking a \$500K ARR enterprise deal, and joined Freshworks with the team on acquisition.

EDUCATION

Bachelor of Engineering (B.E.), Computer Science

2011 – 2015

K.L.N. College of Information Technology (Affiliated with Anna University), Tamil Nadu, India

LANGUAGES

English: Fluent (Professional Working Proficiency) | **Tamil:** Native | **Hindi:** Intermediate